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JEREMY THALLER

Brooklyn, NY

Senior machine learning engineer & data engineer with two MS degrees in physics / materials science and 5+ years shipping production ML systems for performance-critical products in fintech and supply-chain/logistics. Deep experience designing decision engines and high-throughput data/feature pipelines, working with simulation-heavy and optimization workloads, and partnering with engineers and customers to deploy, monitor, and debug ML in production. Comfortable owning end-to-end delivery from scoping and architecture through modeling, experimentation, and translating results into reusable tools, APIs, and libraries.

EXPERIENCE

VEHO | Last-mile delivery company powered by a dynamic driver marketplace
Senior Full-Stack Data Scientist, Marketplace | New York, NY (Remote) [May 2023 – Feb. 2026]

- Led a major overhaul of the route-generation decision engine, coordinating cross-functional changes across engineering and operations; achieved record efficiency and cost savings, reducing cost by $0.17 per package (~$2M/year), validated through pre/post analysis and large-scale simulation-based evaluation.
- Built a distributed computing platform for **large scale route simulations**, including ETL pipelines, Ray based simulations on AWS EC2, and integration of results into Redshift/DBT backed dashboards to continuously evaluate routing algorithms, guard against regressions, and support "what if" analysis for product decisions.
- Led **end-to-end development of a delivery offer recommendation system** (XGBoost) from feature engineering and training through deployment and monitoring, reducing offer claim time and improving marketplace responsiveness; defined offline/online success metrics and evaluated impact via controlled online experiments.
- Redesigned a MILP-based offer–route **matching algorithm** to optimize for driver-level personalization and quality, reducing misdeliveries by 3% as measured by randomized controlled experiments and causal inference analysis.
- Built and maintained a production-grade internal analytics and decision-support application (Python/Streamlit on ECS/Fargate), enabling non-technical teams to self-serve complex route, cost, and model performance analyses on top of warehouse data.

CURRENT | Financial technology company with no-fee banking, payroll deposits, debit and credit cards
Data Scientist, Product / Core-Banking | New York, NY (Hybrid) [Aug. 2022 – May 2023]

- Built user-level **segmentation and early-indicator models** (k-means/PCA + supervised learning on Vertex AI / BigQuery ML) to predict long-term user value from 30 days of behavior, reducing feedback cycles for product and marketing experiments by 6× and tying outputs directly to engagement and revenue metrics.
- Modeled product-gating scenarios to forecast churn and revenue trade-offs using **SARIMA forecasting and causal impact analysis**; recommendations drove changes that yielded ~$200k/month in cost savings while preserving core health metrics.
- Standardized a company-wide “active customer” metric using Kaplan–Meier survival analysis and instrumented event data, aligning product, marketing, and finance around a single definition and powering downstream monitoring and experimentation.
- Wrote **Airflow DAGs and SQL transformations** to populate daily updating BigQuery tables backing Looker dashboards for product and growth teams, and contributed to the later DBT migration to improve reliability and scalability of analytics pipelines.

CELSIUS NETWORK | Centralized finance crypto company providing various financial services, formerly with $30B AUM
Data Analyst, Growth and Marketing | New York, NY (Remote) [Sept. 2021 – Aug. 2022]

- Developed supervised ML models (gradient boosting) to score and prioritize high-value prospects, expanding the high-net-worth prospect pool by 10× and driving ~$50B in potential platform growth; evaluated models with precision/recall, ROC-AUC, and calibration checks.
- Designed and deployed an **anomaly detection system** to alert on fraud and platform violations, including metric definitions, thresholding, and monitoring to balance false positives against missed incidents.
- **Built and maintained internal Python packages** to standardize recurring SQL queries and transformations, improving reliability, testability, and speed for analytics workflows across teams.

BROOKHAVEN NATIONAL LABORATORY | Structure and Dynamics of Applied Nanomaterials
MS Thesis Researcher in Deep Learning | Upton, NY [Feb. 2021 – Sept. 2021]

- Used **TensorFlow and convolutional neural networks** to predict absorption spectra disorder, reducing required simulation data by 90% and cutting compute time by 50x via a novel multinomial Gaussian statistical approach.
- Applied **transfer learning** to map simulation-trained models onto experimental datasets, demonstrating generalization from synthetic to real-world physics data and reducing reliance on expensive numerical simulations.

EDUCATION

LUDWIG MAXIMILIANS UNIVERSITÄT & TECHNISCHE UNIVERSITÄT MÜNCHEN | Munich, Germany | 2019 – 2021
M.S. Materials Science and Engineering

ADAM MICKIEWICZ UNIVERSITY | Poznań, Poland | 2019 – 2021
M.S. Computational Physics

WILLIAMS COLLEGE | Williamstown, MA | 2015 – 2019
B.A. Physics, Honors | Sigma Xi Honor Society | Varsity Track & Field Captain

SKILLS AND TOOLS

Languages: Python (7+ yrs), SQL (7+ years), Java, MATLAB, R, Julia
ML & Analytics: XGBoost, scikit-learn, PyTorch, TensorFlow; recommendation/ranking, anomaly/fraud detection, segmentation, forecasting, survival analysis; model calibration and monitoring; offline vs online evaluation, A/B and switchback experiments, causal inference
Data & Infrastructure: Snowflake, Redshift, BigQuery, Databricks, PySpark, Dask, DBT, Apache Airflow, AWS (EC2, Fargate, S3, Athena), Ray, Docker